

MAXIMILIAN COMFERE

New York, NY | (507) 250-6563 | mkc2182@columbia.edu

EDUCATION

Columbia University, Fu School of Engineering and Applied Science
Bachelor of Science, Computer Engineering

New York, NY
Expected May 2026

PROFESSIONAL EXPERIENCE

Department of Physiology and Biomedical Engineering, Mayo Clinic
Engineering Co-op Intern

Rochester, MN
June 2025 – August 2025

- Developed a **multi-classifier ML pipeline** and CNN architectures in **PyTorch**, achieving **94% accuracy** in classifying photoplethysmography (PPG) waveforms on a 0–5 quality scale.
- Proposed a novel construction: Large Physiological Signal Models (LPSMs), leveraging transformer architectures and Llama 3.1 to capture long-range waveform dependencies and integrate temporally aligned HER textual context.
- Built a Python package for waveform annotation with a **FastAPI backend**, enabling thread-safe multi-user labeling and real-time training on up to **12 concurrent data streams**.

Columbia's Mortimer B. Zuckerman Mind, Brain and Behavior Institute
Undergraduate Software Lab Assistant

New York, NY
September 2023 – August 2024

- Implemented MoSeq, a **motion sequence machine learning model**, to observe and then map behavioral characteristics of Naked Mole Rats to specific castes within the colony.
- Designed large-scale **transition matrices** using Python to create visual representations of behavioral sequences.
- Modeled a tunnel game in Python to efficiently pair all Naked Mole Rats against each other to determine a dominance hierarchy for the colony.

Center for Digital Health, Mayo Clinic
Software Engineer Intern

Rochester, MN
June 2023 – August 2023

- Automated synthetic test data generation for a myelodysplastic syndrome (MDS) risk assessment pipeline using **Python, Pandas**, and OpenAI's API, reducing manual **QA time by ~40%**.
- Built a secure internal team portal with **React, TypeScript**, and **Azure App Services**, centralizing workflows, documentation, and model outputs.
- Designed and tested front-end mechanics for a clinical risk calculator, integrating **RESTful APIs, SQL Server streams**, and **Azure cloud** services to deliver real-time predictions.
- Collaborated with data scientists to deploy a machine learning model that analyzed structured EHR and laboratory data to support clinician decision-making at point of care.

EXTRACURRICULARS

Lion Health, Columbia University
Head of Software Engineering

New York, NY
April 2024 – present

- **Lead 35 software engineers** who work in cross-disciplinary teams made of clinicians and hardware engineers to build, deploy and then present a range of biomedical solutions.
- Built a sleep disorder app that uses a **Core ML Model** fine-tuned to wake up patients experiencing involuntary manic and potentially dangerous episodes during sleep.

Track and field, Columbia University

New York, NY
September 2022 - present

- **Honors:** *NCAA USTFCCCA Division 1 All-Academic Team 2023*

SKILLS & INTERESTS

Computer: Java, C/C++, JavaScript, Swift, Python, React, Node, Swift, SQL, PyTorch, MATLAB, Typescript, Verilog, CSS, HTML

Language: English (native), German (advanced)

Interests: Writing, Premier League Soccer, Cycling, Backpacking, Cosmology